

DAY 27 / 30

SUMMIT RIDGE · ML ENGINEERING

Deploying models: APIs, latency, and scaling

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A trained model, sitting on your laptop, helps nobody.

Deployment is the process of turning it into something other systems and users can actually call — and it introduces a whole new set of engineering constraints that have nothing to do with model accuracy.

- ◆ Serving via an API — the model gets wrapped behind an endpoint that accepts input and returns predictions, so any application can call it like any other web service.
- ◆ Latency — how fast is fast enough? A fraud check blocking a payment needs an answer in milliseconds. A report generated overnight can take minutes. The acceptable latency shapes which model size and hardware you can even consider.
- ◆ Scaling — one request is easy. Ten thousand simultaneous requests requires load balancing, request queuing, and often multiple model replicas running in parallel — plus a plan for what happens when demand spikes past capacity.
- ◆ Cost — every inference call (Day 7) consumes real compute. At scale, inference cost, not training cost, usually dominates the long-run budget of an AI product.

“The mindset shift from "building a model" to "deploying a model": accuracy gets you in the door, but latency, reliability, and cost determine whether the product actually survives contact with real users.”

Follow along – Day 28 drops tomorrow.

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